Quantum Universe EXPO
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– Quantum Universe EXPO series –

Group9 Kyu-bit

Optimizing Crop Spacing Based on Soil Conditions

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Background

- Korea’s Specialized Climate/Geographical Characteristics
  ex) leaned land, few plains,
  surrounded by woods or mountains

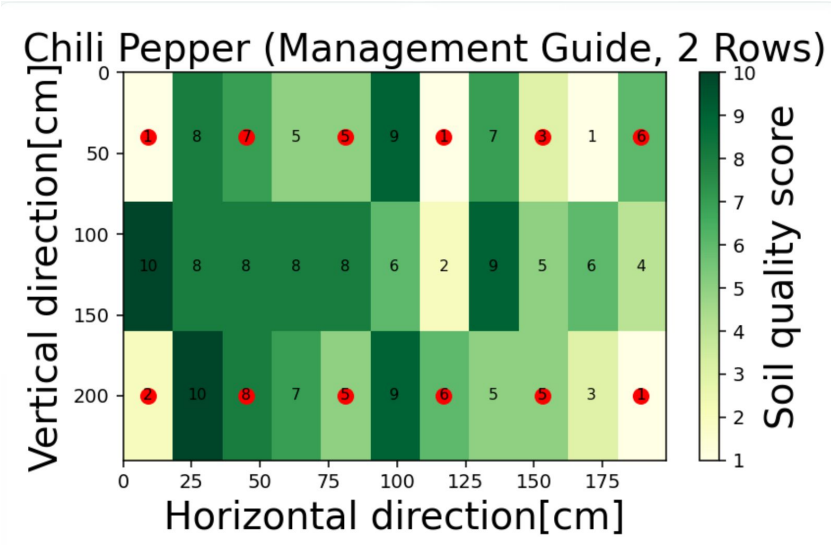
Background

- Few Capital Intensive Farms
  difficulties in
  economics of scale, introduction of new machinery
  →need to use limited areas efficiently

←family farm,
common form of korean farms

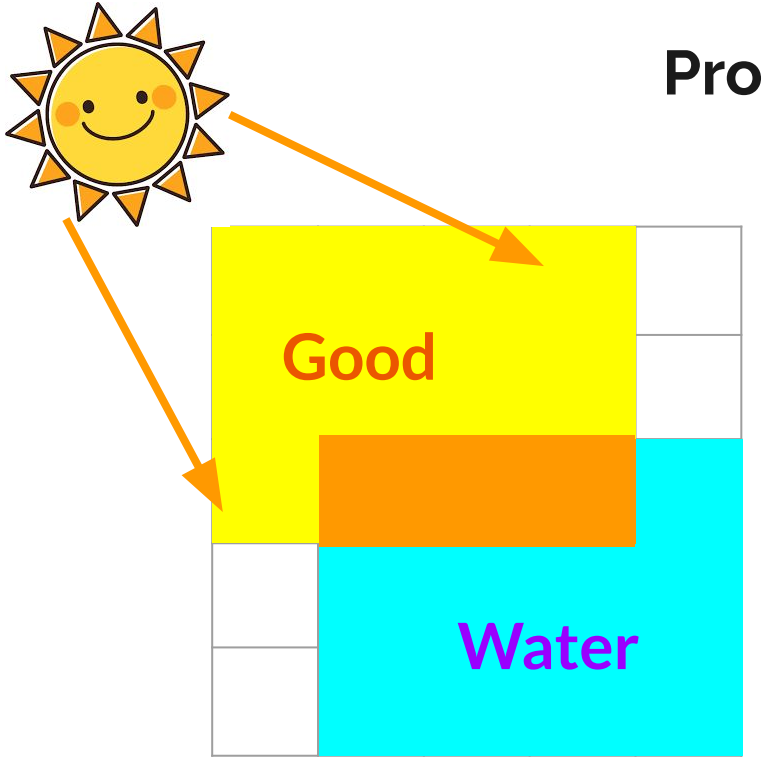
corporate farm,→
capital intensive

Problem Setting



- Each land grid is evaluated and assigned a quality score based on environmental conditions.
- each crop type has a minimum required land size as a constraint.

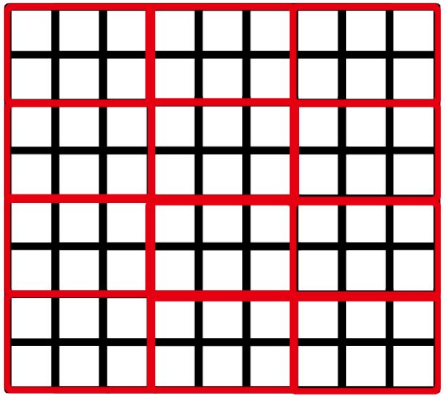
Problem Setting



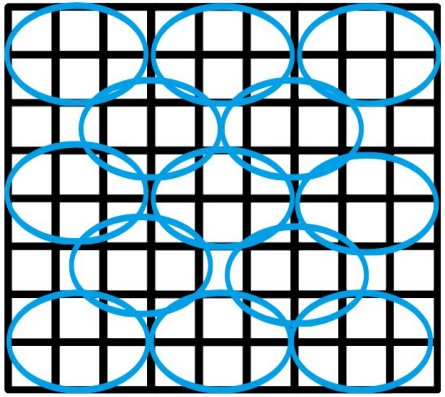
7	7	7	7	1
7	7	7	7	1
7	10	10	10	4
1	4	4	4	4
1	4	4	4	4

Problem Setting

The shape of required land size



Rectangle



Ellipse

Problem Setting

0	1	0	1	0
1	0	0	0	1
0	1	1	0	0
1	0	0	0	1
0	1	0	1	0

$$x_i = \begin{cases} 1 & \text{if area } i \text{ is planted} \\ 0 & \text{otherwise (unused)} \end{cases} \quad \text{for } i = 0, \ldots, N-1$$

$$\mathbf{x} \in \{0,1\}^N, \quad \text{where } N = H \times W$$

$$H_{\text{total}} = H_{\text{fertility}} + H_{\text{adjacency}} + H_{\text{count}}$$

- $H_{\text{fertility}} = -\sum_{i=1}^{N} S_i x_i$
- $H_{\text{adjacency}} = P_{\text{adj}} \sum_{(i,j)\in\text{Adj}} x_i x_j$
- $H_{\text{count}} = P_{\text{cnt}} \left(\sum_{i=1}^{N} x_i - K\right)^2$

$$\min H(\mathbf{x}) = \underbrace{-\sum_{i=1}^{N} S_i x_i}_{\text{Fertility}} + \underbrace{P_{adj} \sum_{(i,j)\in Adj} x_i x_j}_{\text{Constraint: Adjacency}} + \underbrace{P_{cnt} \left(\sum_{i=1}^{N} x_i - K\right)^2}_{\text{Constraint: Count}}$$

S_{i}:

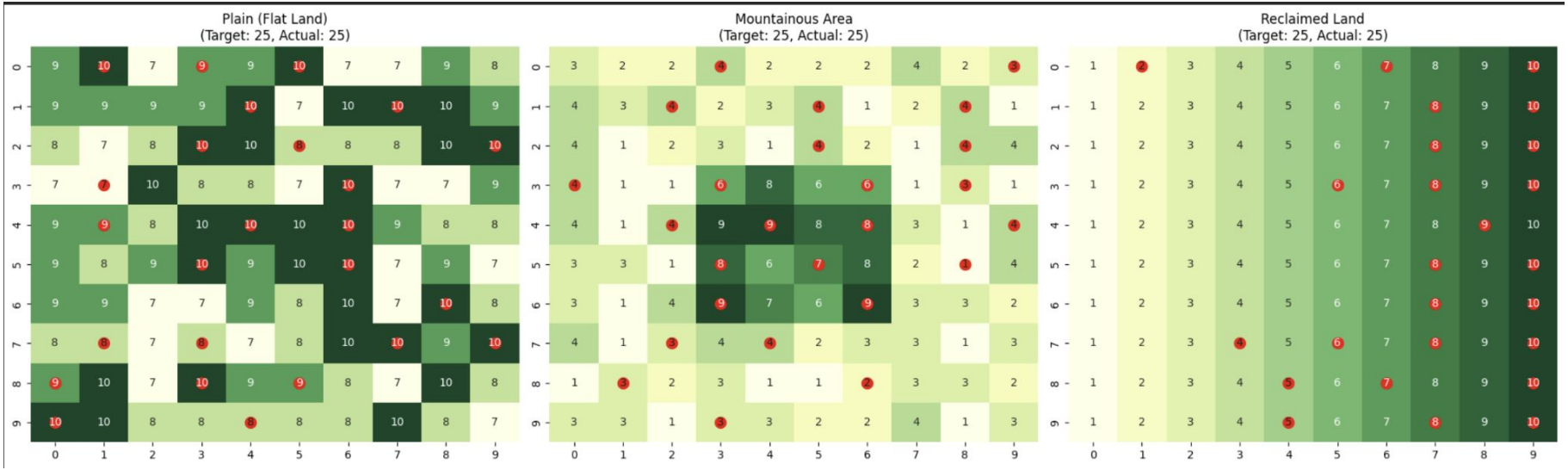
Here, sharing the application！

Kyu–AOS

MAKE FARMING EASIER

Result (Rectangle)

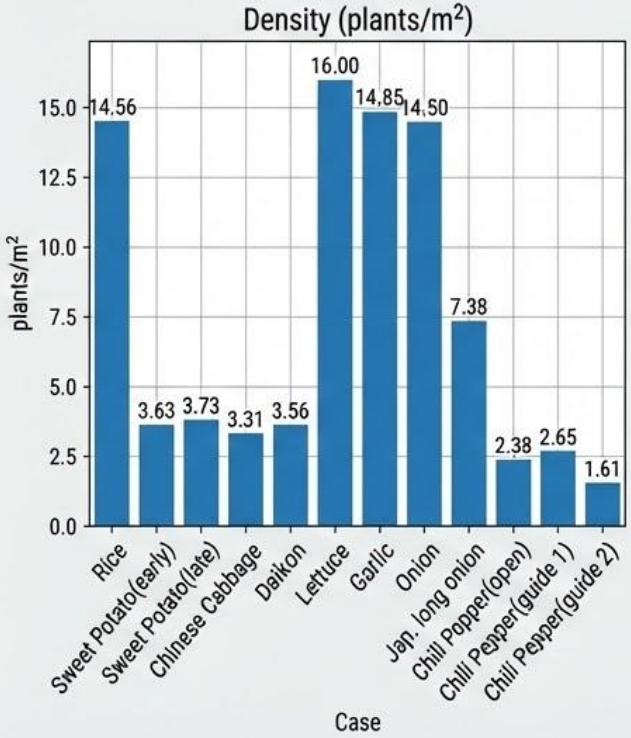
Geographical Diversity(using rectangle)

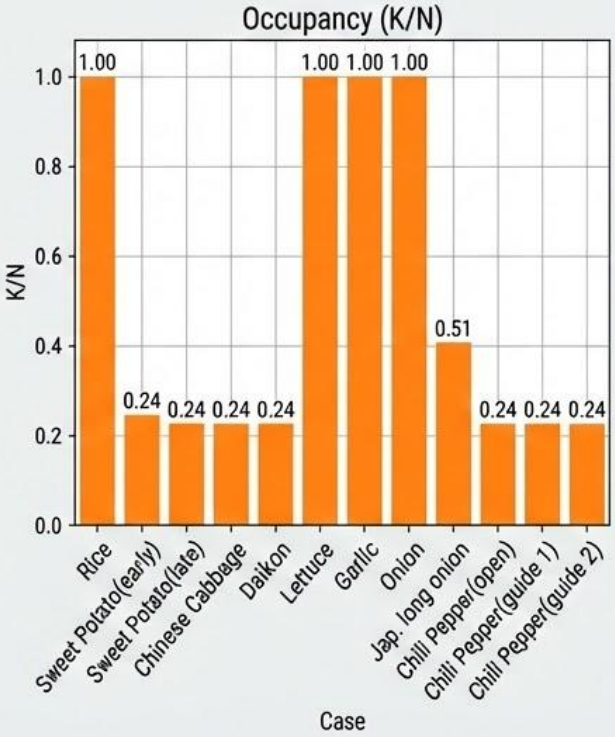


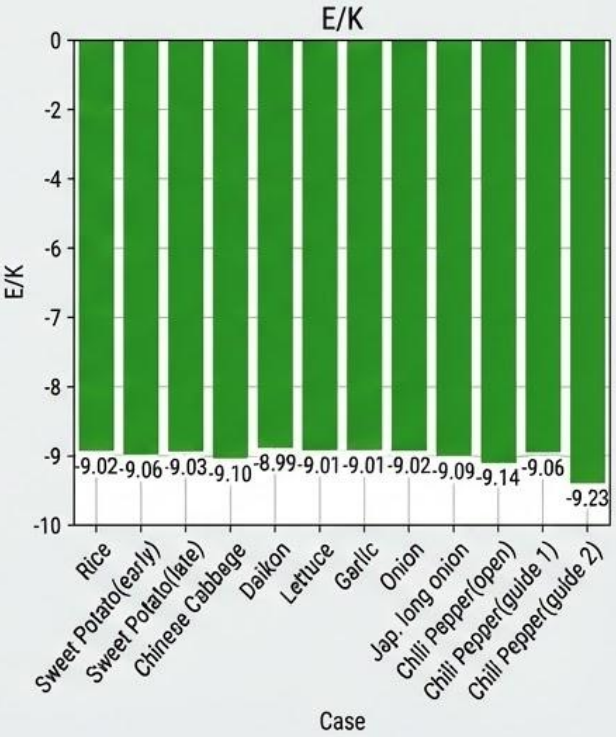
Model 1 (Square Case) Results: Normalized Data

Crop/Case	Grid	N	K(actual)	K/N	Density (plants/m^2)	Best E	E/K
Rice	25×52	1300	1456	**1.120**	14.56	-13128.899	-9.0171
Sweet Potato (early/normal)	20×76	1520	363	0.2388	3.63	-3288.667	-9.0597
Sweet Potato (late season)	22×71	1562	373	0.2388	3.73	-3368.444	-9.0307
Chinese Cabbage	28×50	1400	331	0.2364	3.31	-3011.556	-9.0984
Daikon/Radish	27×55	1485	356	0.2397	3.56	-3201.111	-8.9919
Lettuce	40×40	1600	1600	**1.000**	16.00	-14413.333	-9.0083
Garlic	27×55	1485	1485	**1.000**	14.85	-13384.889	-9.0134
Onion	25×58	1450	1450	**1.000**	14.50	-13078.889	-9.0199
Japanese long onion	13×111	1443	738	0.5114	7.38	-6711.778	-9.0946
Chili Pepper (open field, single row)	18×55	990	238	0.2404	2.38	-2174.222	-9.1354
Chili Pepper (mgmt guide, 1 row)	20×55	1100	265	0.2409	2.65	-2400.222	-9.0574
Chili Pepper (mgmt guide, 2 row)	12×55	660	161	0.2439	1.61	-1486.667	**-9.2340**

Model 1 Results: Density, Occupancy, and E/K Comparison

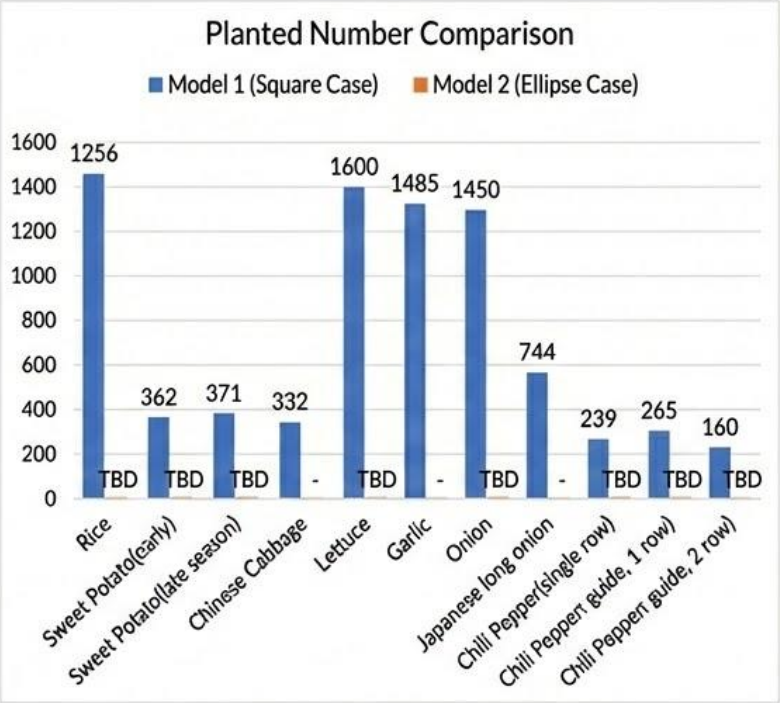






Compare the Results

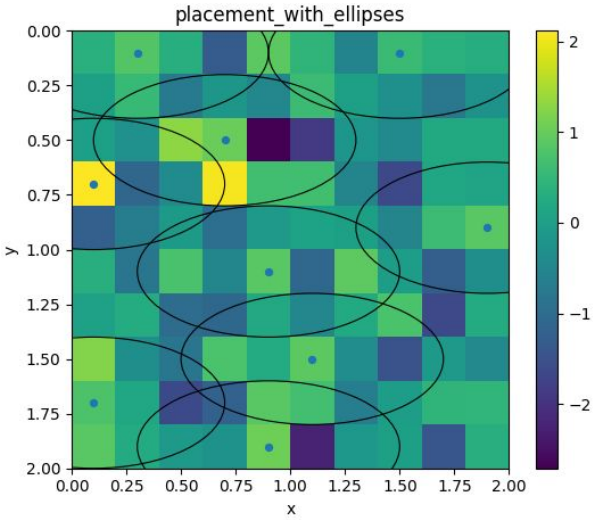
Crop	Model 1 (Square Case) - Planted Number
Rice	1256
Sweet Potato(early nomal)	362
Sweet Potato(late season)	371
Chinese Cabbage	332
Daikon/Radish	356
Lettuce	1600
Garlic	1485
Onion	1450
Japanese long onion	744
Chili Pepper(open field, single row)	239
Chili Pepper(management guide, 1 row)	265
Chili Pepper(management guide, 2 row)	160



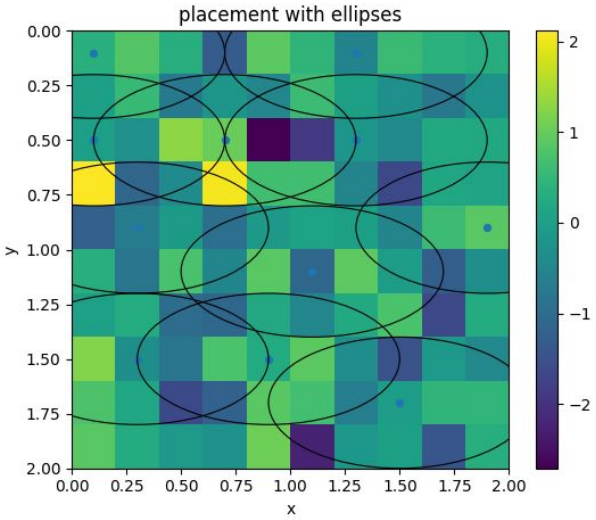
Result (ellipse-based)

Problem Setting : rows = 10, cols = 10

Simulated Annealing



Quantum Annealing



Further adjustment of constraint parameters is required.

Conclusion and Future Work!

Conclusion

This implementation optimized crop placement by demonstrating the best arrangement for each crop on grid-cut plots.

For crop spacing, we devised two patterns: one based on squares and another based on ellipses.

- The square-based method achieved effective placement.
- The ellipse-based method had overlapping sections that require improvement.

Future

We aim to compare how much farmland can be covered using the square pattern versus the ellipse pattern, highlighting the differences between them.